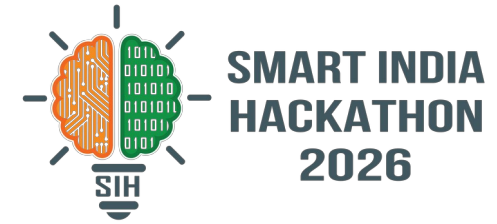


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID** – 26169
- **Problem Statement Title** – AI-Based Virtual Camera Tracking System for Coarse Alignment of Mobile FSOC Terminals
- **Theme** – Smart Automation
- **PS Category** – Software
- **Team ID** – —
- **Team Name** – ZeroDrift

The problem in one line: a mobile laser (FSOC) terminal must find, verify and continuously point at its partner's beacon with hair-thin accuracy — ISRO asks for this coarse-alignment brain, proven entirely in software.



IDEA TITLE

- ❖ *Proposed Solution (Describe your Idea/Solution/Prototype)*
- *Detailed explanation of the proposed solution*
- *How it addresses the problem*
- *Innovation and uniqueness of the solution*

THE PROBLEM why pointing is the hard part

μrad

Laser beams are hair-thin. Both terminals must stay pointed at each other to micro-radians, on platforms that move and shake.

Kaymak et al., IEEE Commun. Surveys & Tutorials 20(2), 2018

100 Gbps

The record 100 Gbps link to a moving drone stayed up only because a camera tracked the target (10 Hz vision + 200 Hz tip/tilt).

Walsh et al., Scientific Reports 12, 2022

1/5 cost

Free-space optics costs about one-fifth of laying fibre and deploys in hours — India's last-mile and disaster links need it mobile.

TEC / DoT, White Paper on Free Space Optics

ISRO PS26169 asks for the coarse-alignment brain:

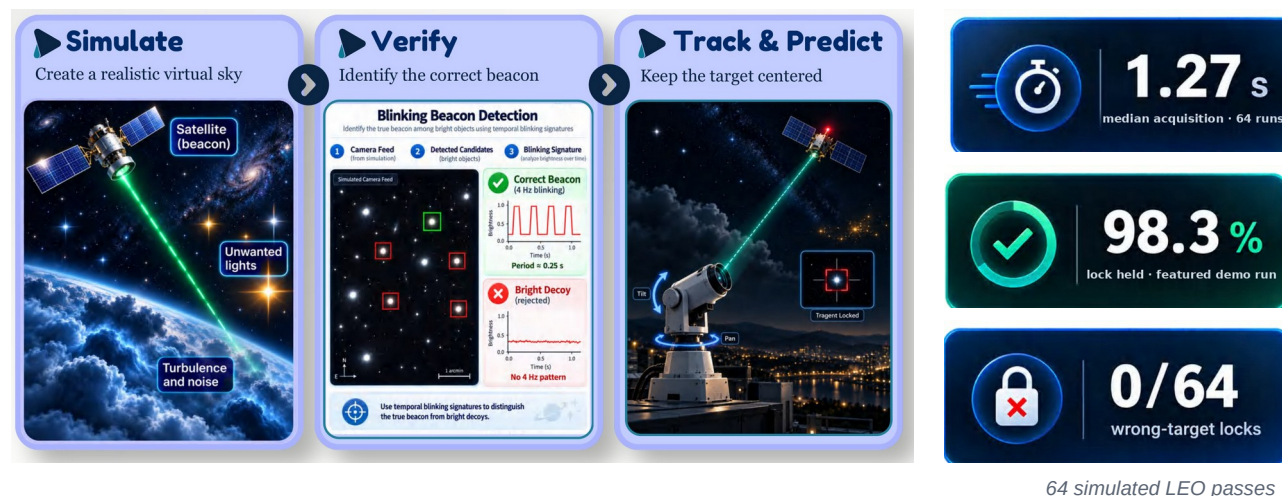
find the partner's beacon among stars, glints and decoys, prove it is the right one, and hold it centred — in software.

≤ 2 s acquire

≤ 10 px error

≥ 20 FPS

OUR SOLUTION — ZERODRIFT the eyes and neck of a laser terminal



01

Identity, not brightness

Locks only on the 4 Hz blink. Lamps, stars, glints and wrong-rate strobes are rejected.

02

Built and running

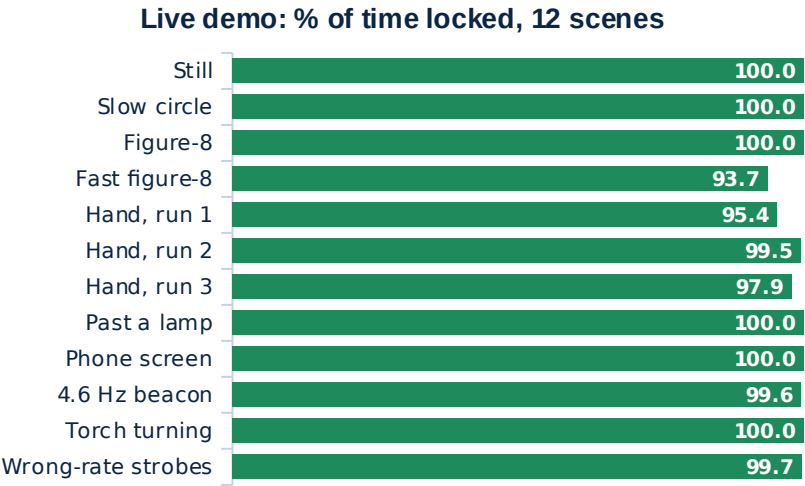
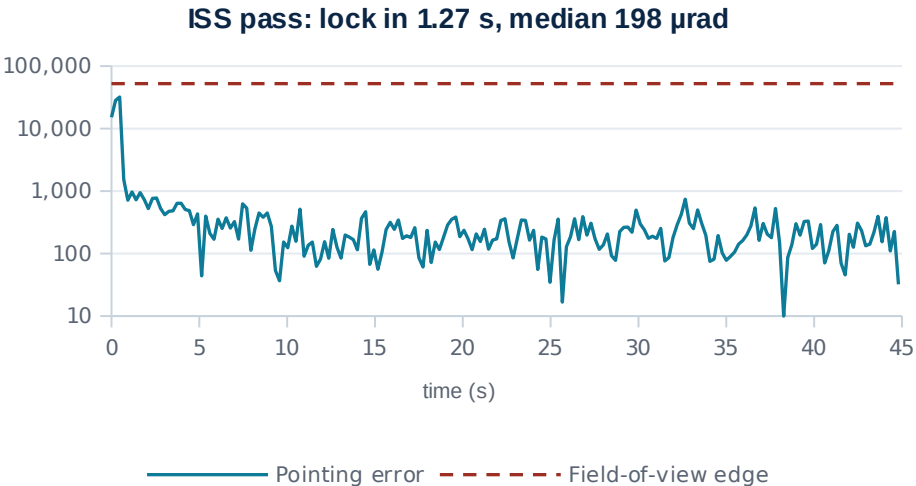
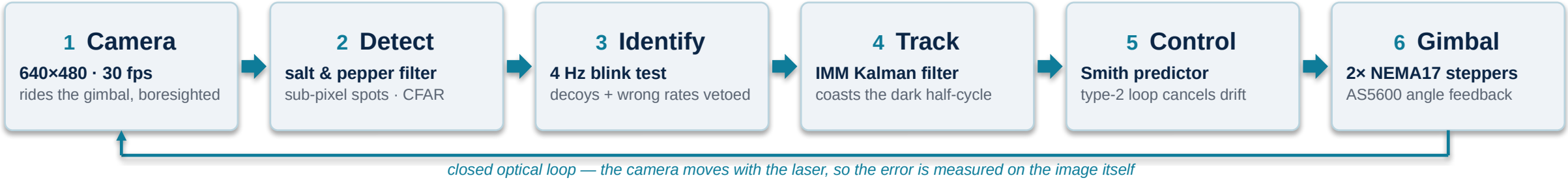
Live webcam demo, MK1 servo rig, MK2 stepper terminal — ₹6,709 bill of materials.

03

Measured, not claimed

250 passing tests. Every number on these slides regenerates from one command.

- Technologies to be used (e.g. programming languages, frameworks, hardware)
- Methodology and process for implementation (Flow Charts/Images/ working prototype)



BUILT WITH

Engine
Python · NumPy · OpenCV

Web demo
JavaScript · three.js · Web Serial

Hardware
Arduino Nano · A4988 · AS5600

Proof
pytest · 250 tests · GitHub Actions

Acquisition
0.55–1.1 s in the live demo

- Analysis of the feasibility of the idea
- Potential challenges and risks



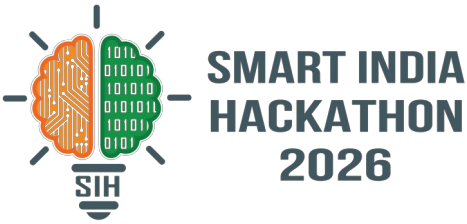
64 runs, our LEO scenarios → never a wrong lock, never a failed acquisition.



Reproducible: every number regenerates from one command.



IMPACT AND BENEFITS



- Potential impact on the target audience
- Benefits of the solution (social, economic, environmental, etc.)

ECONOMIC FEASIBILITY — the arithmetic that removes the cost barrier

CAPITAL COST TO START DEVELOPING

Hardware PAT test bench

₹15–50 lakh

Machine-vision camera + encoder gimbal + fast steering mirror + beacon optics + isolation table + turbulence cell

ZeroDrift

₹0

Runs CPU-only on a laptop the team already owns — no GPU, no cloud, no purchased dataset

₹0.22

Electricity for a complete 64-run validation campaign

0.0279 kWh measured × ₹8/kWh. Per single run: ₹0.0035

207×

Faster than a hardware bench over the same 64-condition sweep

37 minutes measured, against ~16 working days of lab reconfiguration

₹15 cr

Hardware needed to equip 100 researchers independently

ZeroDrift equips the same hundred for nothing

₹15 crore vs ₹0

STRATEGIC IMPACT

- Advances India’s next-generation space & defence communications
- Sovereign FSOC algorithms
- Less imported hardware

RESEARCH & EDUCATION

- Any student or ISRO lab can develop, test and compare PAT algorithms instantly
- Reproducible scenarios, honest ground-truth scoring, open source

ECONOMIC IMPACT

- A laptop replaces lakhs of rupees of cameras, gimbals and optics
- Software-based testing
- Zero recurring cost

SCALABLE & GREEN

- Same engine covers satellite, UAV, ground and maritime links
- Pure software- no consumables, no lab energy, no e-waste in the development cycle.

Target audience

ISRO

DRDO Teams

IITs & engineering colleges

space-tech startups

Optical Link Close Rate

the number that matters

ZeroDrift coarse + modelled fine stage

99.3%

Coarse stage alone — the gap we close

14.7%

same tracked ISS pass, identical channel · mean margin 3.1 dB · docs/technical_report.md

PS26169 compliance: their table, our values

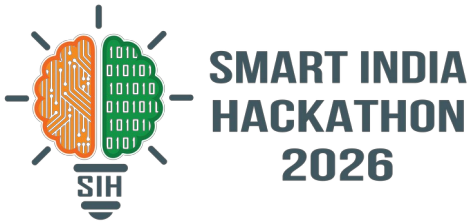
PS26169 PARAMETER	SPECIFIED	ZERODRIFT	
Camera resolution	640 x 480	640 x 480	✓
Camera FOV	4° x 3° default	4° x 3°	✓
Camera update rate	≥ 30 Hz	30 Hz	✓
Max pan / tilt speed	5-10°/s, default 5	5°/s	✓
Target shape / size	square, 10 x 10 px	square, 10 x 10 px	✓
Motion patterns	≥ 4, incl. Figure-of-8	7, incl. Figure-of-8	✓
Image noise	S&P ~10%, Gaussian, Poisson	all three, S&P at 10%	✓
Camera jitter	± 20 px / frame	19.8 px p99, measured	✓
Platform motion	± 20 px / frame, linear	linear drift, 6 px / frame	✓
Atmospheric conditions	Clear / Haze / Fog / Rain / Low light	all five	✓
MP4 input, bypassing PTZ	required (Benchmark-2)	--video, working	✓
Acquisition time	≤ 2 s	25.3 s → 3.53 s — open	✗
Tracking error (median)	≤ 10 px	2,367 → 170 px — open	✗
Processing speed	≥ 20 FPS	9.2 → 14.7 FPS — open	✗

The three open rows are open, not footnoted. Arrows are before → after our own fix, both measured with the full specified disturbance present. A constant-velocity platform drift is a ramp, and a PD loop has finite steady-state error to a ramp — wrong loop order, not bad tuning. Adding a second integrator (type 2, leaky) cut acquisition 7x and median error 14x. It is implemented and shipped; it is not yet enough.

Every figure regenerates from `scenarios/ps26169_benchmark.yaml`, which encodes their table and nothing else.



RESEARCH AND REFERENCES



• *Details / Links of the reference and research work*

Research foundations

Kaymak et al., “A Survey on Acquisition, Tracking and Pointing Mechanisms for Mobile FSO,” IEEE Comm. Surveys & Tutorials, 2018. [\[link\]](#)

Kaushal & Kaddoum, “Optical Communication in Space: Challenges and Mitigation Techniques,” IEEE CS&T, 2017. [\[link\]](#)

Blackman & Popoli, Design and Analysis of Modern Tracking Systems — IMM filtering, CFAR detection. [\[link\]](#)

Bar-Shalom et al., Estimation with Applications to Tracking and Navigation — Kalman/IMM theory. [\[link\]](#)

Smith, O.J.M., “Closer Control of Loops with Dead Time” — the Smith predictor, 1957. [\[link\]](#)

Vallado & Crawford, “SGP4 Orbit Determination” — AIAA 2008; real ISS TLE propagation. [\[link\]](#)

Andrews & Phillips, Laser Beam Propagation through Random Media — turbulence & scintillation. [\[link\]](#)

Our work — open and verifiable

Repository: github.com/pranshukesavmishra/SIH-2026

Live demo (replay console): zerodrift-fsoc-pat.netlify.app

Inside: 5,500+ documented lines · 250 passing tests · 6 validated scenarios · Monte-Carlo logs · technical report · user manual · demo video.

Reproducible: every figure regenerates from one command; every random draw is logged and replayable.

Compared with existing approaches

Compared with existing approaches			
Feature	ZERO+DRIFT	MATLAB sim	Traditional
Zero hardware cost No special equipment required	✓	✓	✗
Real-time closed loop + live console See it run. Live.	✓	✗	✓
Decoy & blink-signature discrimination Robust against spoofing	✓	✗	✗
Open source + one-command reproducible Run with a single command	✓	✗	✗

Why ZeroDrift wins on this problem

- ✓ All six mandatory deliverables already exist today
- ✓ Judged numbers, honestly measured against hidden ground truth
- ✓ AI where it earns its place — benchmarked against classical methods

